Lecture 1 — Membrane transport

Cell Biology (demo course) · Week 1

1.1 The plasma membrane

The plasma membrane is a phospholipid bilayer with embedded proteins (the fluid mosaic model). Its hydrophobic core lets small non-polar molecules (O_2, CO_2) pass, but blocks ions and large polar molecules such as glucose. The membrane is therefore **selectively permeable**.

1.2 Passive transport

Passive transport needs no energy from the cell: substances move **down** their concentration gradient, from high to low concentration, until equilibrium.

- **Simple diffusion** — directly through the bilayer (O_2, CO_2, steroid hormones).
- **Facilitated diffusion** — through channel proteins or carrier proteins (glucose via GLUT transporters, ions via ion channels). Still passive: no ATP is used.

The rate of diffusion increases with a steeper gradient, higher temperature and a larger surface area, and decreases with a longer diffusion distance.

1.3 Osmosis

Osmosis is the passive movement of **water** across a selectively permeable membrane, from a region of lower solute concentration to a region of higher solute concentration. Many cells speed it up with water channels called **aquaporins**.

Tonicity describes the solution outside the cell, compared with the cytoplasm:

Outside solution	**Water moves**	**Animal cell (red blood cell)**	**Plant cell**
Hypotonic (less solute)	into the cell	swells, may burst (lysis)	turgid — the cell wall prevents bursting
Isotonic (equal)	no net movement	normal	flaccid
Hypertonic (more solute)	out of the cell	shrinks (crenation)	plasmolysis — the membrane pulls away from the wall

> **Typical exam mistake:** water does not "follow the higher water concentration". It moves towards the side with the *higher solute* concentration (lower water potential).

1.4 Active transport

Active transport moves substances **against** their concentration gradient, using energy, usually from ATP. It needs specific carrier proteins called pumps.

Key example — the **sodium–potassium pump** (Na^+/K^+-ATPase): for each ATP hydrolysed it moves **3 Na^+ out** of the cell and **2 K^+ in**. This keeps the inside of the cell negative and maintains the resting membrane potential of neurons.

1.5 Bulk transport

- **Endocytosis** — the membrane folds inwards to take material in: phagocytosis ("cell eating", e.g. a white blood cell engulfing bacteria) and pinocytosis ("cell drinking").
- **Exocytosis** — vesicles fuse with the membrane to release their contents (e.g. neurotransmitters, hormones).

Both use ATP, so they count as active processes.

Summary

Passive (diffusion, facilitated diffusion, osmosis): down the gradient, no ATP. Active (pumps, endo- and exocytosis): against the gradient or in bulk, needs ATP.